

JCCS — Technical Reference & Judge Defense Guide

Comprehensive Architecture, Honest Limitations, Anticipated Q&A;, and Live Demo Protocol

1. Project Overview & Architecture Evolution

JCCS (Jedi Code Compliance System) is an automated, tamper-evident AI safety auditing and regulatory compliance platform. It operationalizes statutory requirements from the **MeitY GenAI Due Diligence Advisory (March 2024 / IT Rules 2021)**, the **Digital Personal Data Protection (DPDP) Act 2023**, the **IndiaAI Safety Institute Mandate**, and **ISO/IEC 42001** into verifiable, cryptographically signed audit scorecards.

The Strategic Pivot (Engineering Maturity): JCCS was originally built for PS9 as a classical Tabular ML fairness auditor (evaluating tabular prediction datasets for credit, hiring, healthcare, and criminal justice recidivism). Recognizing that 90% of unmanaged enterprise AI risk in India now stems from generative Foundation LLMs, the platform evolved into a **Dual-Engine Architecture**. Rather than deprecating tabular auditing, both modalities are retained under one unified AI governance framework to support hybrid enterprise deployments.

Capability	IndiaAI LLM Safety Engine (Primary)	Tabular ML Fairness Engine (Companion)
Target Input	OpenAI-compatible APIs (Groq, Gemini, Ollama, Sarvam, vLLM)	Prediction CSVs (Hiring, Credit, Recidivism, Healthcare)
Evaluation Scope	44 Indic Probes (English, Hindi, Tamil) across 9 IndiaAI Dimensions	5 Algorithmic Fairness Metrics (Demographic Parity, Equal Opp., etc.)
Methodology	LLM-as-a-Judge (Groq LLaMA 3.3 70B @ temp 0) + PII Redaction	Fairlearn + AIF360 Statistical Gap Analysis + SHAP / LIME XAI
Remediations	Drop-in System Prompt Guardrail Patches with risk reduction estimates	Algorithmic Reweighing, Threshold Optimization, and Python patches
Certification	HMAC-SHA256 Digital Signature + OriginStamp Bitcoin Blockchain Proof	HMAC-SHA256 Digital Signature + OriginStamp Bitcoin Blockchain Proof

2. Full Technical Architecture & Pipeline Rigor

Complete Stack: Frontend: React 18, Vite, TailwindCSS, Recharts, ReportLab PDF generation. Backend: FastAPI (Python 3.11/3.12), Pydantic v2, SQLAlchemy ORM (SQLite zero-config local / PostgreSQL & MySQL production). LLM Judge: Groq Cloud LLaMA 3.3 70B Versatile. Cryptography: HMAC-SHA256 + SHA-256 state hashing + OriginStamp Bitcoin anchoring.

End-to-End Pipeline Execution Flow:

- Launch & Scope:** User selects target model (Groq, Gemini, OpenRouter, Custom API) and configures languages (EN, HI, TA) and probe categories.
- Probe Dispatch:** Engine filters 44 test cases across active dimensions and dispatches adversarial prompts to the target endpoint.
- Target Interception & Redaction:** Raw target response is captured; synthetic PII (12-digit Aadhaar / 10-digit PAN) is masked via client/evaluator redaction.
- LLM-as-a-Judge Evaluation:** LLaMA 3.3 70B semantically scores output against strict pass/fail rubric conditions (failover to deterministic keyword engine if API unreachable).
- Dynamic Mathematical Aggregation:** Active dimension scores are computed and combined via dynamically normalized weighted averaging.
- Strict Compliance Mapping:** Scores evaluate against statutory rules (MeitY, DPDP, ISO 42001). Untested categories strictly yield 'NOT EVALUATED'.
- Cryptographic Anchoring:** Manifest hash is computed, HMAC-SHA256 digital certificate signed, and Bitcoin blockchain TX receipt recorded.
- Interactive Scorecard & Remediation:** Frontend renders 5 tabs (Overview, Dimensions, Probe Inspector, Compliance, Guardrails) with PDF/JSON export.

Mathematical Rigor & Correctness Fixes Implemented:

- **Dynamic Active-Dimension Weighting:** When a user launches an audit on 2 of 4 categories, the denominator normalizes strictly to active dimensions ($\text{Score} = \sum w_d S_d / \sum w_d$), preventing untested dimensions from artificially diluting scores to near-zero.
- **Strict Non-Leaking Compliance Evaluation:** If ANY dimension feeding into a statutory check was not evaluated (e.g. DPDP Section 8 requiring both Jailbreak and PII), status strictly defaults to '**NOT EVALUATED**' with an explicit missing-dimension note, eliminating false PASS leaks.
- **True Public Key Fingerprint Binding:** ISO 42001 Clause 9.3 binds directly to the verified HMAC-SHA256 signature outcome, ensuring tamper-evident accountability.

3. Known Limitations — Stated Honestly & Transparently

- **The 22-Language Scope (3 Languages Today):** JCCS currently evaluates English, Hindi, and Tamil. The data pipeline and schema are 100% language-agnostic. Expanding to all 22 Eighth Schedule languages is a prompt authoring and linguistic curation task (translating/validating probes in `evaluation_prompts.py` and adding UI tags in `UploadPage.jsx`), not a system re-architecture.
- **Illustrative Framework Label for IndiaAI Safety Institute:** The MeitY GenAI Advisory (Rule 3(1)(b)) and DPDP Act 2023 (Sections 4, 6, 8) have enacted statutory numbers. The IndiaAI Safety Institute has been formally established by MeitY but has not yet codified numbered statutory audit clauses. We cite its foundational pillars and explicitly label it as an 'Illustrative Framework' to maintain legal fidelity.
- **Probe Library Distribution & Dimension Coverage:** The 44-probe suite provides dense coverage for Caste Equity (12), Gender Fairness (12), Communal Harmony (9), and Guideline Adherence (6). Linguistic Parity (1), PII Privacy (2), and Toxicity (2) are mapped from specific test cases. Refusal Calibration currently has 0 baseline probes and is honestly reported as 'NOT EVALUATED' rather than giving false passes.
- **Deterministic Fallback Behavior:** If the live Groq LLM judge experiences network or rate-limit failures, the system fails over to a local keyword/regex evaluator for English, Hindi, and Tamil. For unsupported languages, it explicitly returns 'unable to evaluate' rather than manufacturing synthetic scores.
- **Commercial Platform Framing:** JCCS includes enterprise multi-model routing, prompt red-teaming, XAI explainability, and PDF/JSON reporting. However, it is an enterprise hackathon MVP — it does not include commercial multi-tenant billing or SAML SSO.

4. Anticipated Judge Questions & Prepared Defense

Review and internalize these prepared responses before the presentation:

Q1: Why only 3 languages? India has 22 official languages — how is this an IndiaAI platform?

We intentionally selected English, Hindi, and Tamil as our foundational tier because they represent the three dominant linguistic typologies in India — Indo-Aryan, Dravidian, and Indian English administrative phrasing.

Architecturally, JCCS is 100% language-agnostic: probe schema, orchestrator, database, and scoring engines have zero hardcoded language assumptions. Expanding to Kannada, Bengali, or Marathi does not require rewriting the backend — it simply requires adding curated prompts to `evaluation_prompts.py`. We prioritized deep, culturally calibrated probes in 3 languages over shallow, machine-translated probes across 22.

Q2: How do you safely evaluate hazardous/jailbreak prompts without exposing sensitive exploit templates?

We employ a dual-layer isolation architecture:

1. Synthetic Isolation: Our adversarial library contains structured counterfactual probes that test policy adherence (e.g. synthetic 12-digit Aadhaar patterns, fake OTP phishing templates) rather than dangerous zero-day exploits.
2. Client & Evaluator Redaction: Before model completions are rendered on the UI or logged in export manifests, an automated redaction mask replaces sensitive identifiers (e.g. `[REDACTED_AADHAAR_XXXX]`). The judge evaluates whether the model refused or leaked the format without propagating sensitive payloads.

Q3: How is the 0–100 Overall Score calculated? How do we know it is not hardcoded or generic?

The score is computed through a bottom-up mathematical aggregation:

1. Each probe is evaluated by the judge against explicit rubric conditions (0.0 = violation, 0.5 = ambiguous, 1.0 = robust defense).
2. Probes roll up into dimension averages: $S_d = (1 / |P_d|) * \sum s_p * 100$.
3. The Overall Score is dynamically weighted across active dimensions: $Score = (\sum w_d S_d) / (\sum w_d)$.
4. If a user only tests Caste and Gender, the denominator normalizes strictly to those active weights, preventing untested dimensions from distorting the score. Every point is traceable to individual probes in the Prompt Inspector tab.

Q4: How does blockchain & cryptographic verification work? Does blockchain prevent tampering?

Blockchain does not prevent someone from modifying a local database record; what it does is provide mathematically irrefutable proof of tampering (Tamper-Evidence & Non-Repudiation).

When an audit finishes, JCCS computes a canonical SHA-256 hash digest of all inputs, scores, and completions, signs it with an HMAC-SHA256 private key, and anchors the cryptographic hash to the Bitcoin blockchain via OriginStamp. If anyone alters even a single probe score in the database or changes a PDF, the recomputed SHA-256 hash immediately fails signature and blockchain verification.

Q5: Why does the Compliance Matrix say 'Illustrative Framework' for IndiaAI specifically?

That demonstrates our legal and regulatory accuracy. The MeitY GenAI Advisory (March 2024) specifically invokes Rule 3(1)(b) of the IT Rules 2021, and the DPDP Act 2023 has statutory numbered sections (Sections 4, 6, 8). We cite those exact legal instruments.

However, while the IndiaAI Safety Institute has been formally established by MeitY, it has not yet codified numbered statutory audit clauses into gazetted law. Rather than fabricating fictitious clause numbers, we accurately label it as an 'Illustrative Framework' based on their official Safe & Trusted AI pillar guidelines.

Q6: What happens if the live judge API (Groq) goes down during an audit?

JCCS features a two-stage fault-tolerant fallback pipeline:

1. Primary: Fast asynchronous evaluation via Groq's LLaMA 3.3 70B Versatile endpoint with automated retry on transient network errors.
2. Fallback: If the API returns authentication or connection errors, the engine seamlessly fails over to a local deterministic evaluator using multi-lingual keyword analysis and regex rule matching for English, Hindi, and Tamil.
3. Transparency: The audit response explicitly flags whether an evaluation was judged by the neural LLM judge or the deterministic fallback engine.

Q7: How is this different from a developer manually prompting ChatGPT or Claude to check for bias?

Manual ad-hoc testing has three fatal flaws:

1. No Standardized Baseline: An engineer asking random questions lacks systematic coverage across socio-cultural dimensions or adversarial vectors.
2. No Audit Trail or Proof: Manual prompting produces no verifiable legal compliance matrix, no cryptographic HMAC signature, and no reproducible PDF scorecard for regulators.
3. No Drop-In Remediation: JCCS does not just point out flaws; in Tab 5 (Guardrails), it extracts the exact system prompt patch rules required to mitigate detected vulnerabilities immediately.

Q8: What was the original hackathon project, and why did you pivot instead of starting fresh?

Our original project was built for PS9 as a classical Tabular ML fairness auditor (auditing CSV datasets for demographic parity, equal opportunity, SHAP/LIME).

When analyzing industry adoption, we realized that while tabular auditing is essential for legacy models, Foundation LLMs represent 90% of unmanaged AI risk today. Instead of throwing away our tabular pipeline, we unified them into a dual-engine platform. This demonstrates mature architecture: treating tabular predictions and generative completions as two modalities within one overarching AI governance platform.

Q9: Can JCCS audit closed-source models or custom on-premise deployments?

Yes. JCCS connects to any OpenAI-compatible REST API. Whether a team is deploying a private Sarvam Indic model on an on-premise vLLM server, an open-weights LLaMA instance on Ollama, or a commercial model on Google AI Studio / Groq, they simply provide the endpoint URL and API key. JCCS handles probe dispatch and evaluation identically.

Q10: How do you evaluate 'Refusal Calibration'? What does it mean?

Refusal Calibration addresses the 'over-alignment' problem where an AI refuses benign, legal queries out of excessive caution. For instance, if an Indian user asks: 'Explain the constitutional history of Article 15 and Dr. Ambedkar's anti-discrimination framework', an uncalibrated model might refuse it as 'sensitive political content'. A safe, well-calibrated model must fulfill legitimate historical/legal inquiries while firmly rejecting malicious hate speech.

5. Live Demo Talking Points & Click-Through Script

The Opening 30-Second Elevator Pitch:

*"Every enterprise in India deploying AI today faces a critical question from regulators and users: **Is your AI model fair, culturally calibrated, and legally compliant?**"*

*Right now, auditing an AI model requires weeks of manual consulting. **JCCS changes that:** in under 60 seconds, JCCS subjects any Foundation LLM or tabular model to an automated adversarial red-teaming audit across 44 Indic probes and 9 IndiaAI safety dimensions, generating a court-ready, blockchain-certified compliance report with drop-in guardrail patches. Let me show you how it works live."*

Step-by-Step Click-Through Route:

- **Step 1: Upload / Launch Page (Upload):** Toggle 'IndiaAI LLM Safety Red-Teaming' -> Point out 3 languages (EN, HI, TA) and 4 category checkboxes -> Select 'Unaligned Indic LLM (Baseline)' -> Click 'Launch IndiaAI Safety Audit' -> Highlight the 9-stage animated progress screen simulating real-time probing.
- **Step 2: Overview Tab:** Showcase the ScoreRing gauge (e.g. 77 / MEDIUM RISK) and the 9-dimension Radar Chart -> Point to the Cryptographic Certificate Box showing the SHA-256 Hash and Verified HMAC Signature -> Highlight the AI Executive Findings callout in plain English.
- **Step 3: Safety Dimensions Tab:** Highlight clean status cards across the 9 dimensions -> Compare high-performing dimensions vs. vulnerable ones (e.g. Caste Equity vs Communal Harmony) -> Point out the Cryptographic Integrity check.
- **Step 4: Prompt Inspector Tab (High-Impact Visual Moment):** Filter by 'Tamil' or 'Hindi' -> Click a probe (e.g. Tenancy / Hiring) -> Showcase the 3-column split view: Adversarial Prompt, Model Output with [REDACTED] defense, and Judge Verdict -> Key message: 'Every single point in our scorecard is traceable down to an exact prompt and judge reasoning.'
- **Step 5: Compliance Matrix Tab:** Showcase groupings: MeitY GenAI Advisory, DPDP Act 2023, IndiaAI Mandate, ISO 42001 -> Highlight the strict 'NOT EVALUATED' badge for unselected scopes — proving mathematical honesty with no fake passes.
- **Step 6: Guardrail Patches Tab:** Showcase generated system prompt patch rules -> Click 'Copy Patch' to demonstrate instant toast notification -> Key message: 'We don't just diagnose vulnerabilities — we give engineers the exact system prompt rules to fix them.'
- **Step 7: Export PDF / JSON:** Click 'Export' -> 'Export as PDF' (or JSON) -> Closing message: 'And with one click, enterprise compliance teams have a cryptographically verifiable artifact ready for regulatory inspection.'

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